

Antifungal evaluation of plant essential oils and their major components against toxigenic fungi

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ABSTRACT

Contamination of toxigenic fungi in foods, agricultural commodities and Chinese herbal medicines has posed serious threaten to human and animals' health and safety. Natural antifungal agents from plant essential oils (PEOs) and their active components have been candidates of interest. In this study, the antifungal activities of PEOs from 11 natural plants against three kinds of toxigenic fungi including *Aspergillus flavus*, *Penicillium viridicatum* and *Aspergillus carbonarius* were evaluated and compared through the determination of fungi zone diameter. Results showed that 4 PEOs of *Cinnamomum cassia* Presl, *Litsea cubeba*, *Cymbopogon martini* and *Thymus mongolicus* Ronn expressed stronger inhibitory effects and lower minimal inhibitory concentrations (MICs) against the three kinds of fungi, which would be the ideal antifungal candidates. Then, the chemical compositions of them were analyzed by an optimized gas chromatography-mass spectrometry method with *trans*-cinnamaldehyde and d-limonene, citral and citronellal, *trans*-geraniol, and carvacrol as their main active components, respectively. The six compounds with strong antifungal activity and small MICs on the three kinds of fungi might contribute largely to the inhibitory effects of the 4 selected PEOs containing them. The 4 selected PEOs and their 6 main active components exhibited wide application prospect as the novel natural antifungal alternatives for preventing fungal contamination in different matrices to assure their quality and safety. The use of PEOs with highly-volatile components is environment-friendly, convenient and highly-effective with minimal residues and low cost will get more interest in practice.

1. Introduction

For a long period, the security of foods, agricultural commodities and Chinese herbal medicines (CHMs) due to fungal contamination especially toxigenic fungi such as *Aspergillus flavus*, *Penicillium viridicatum* and *Aspergillus carbonarius* et al. (Asghar et al., 2017; Baquião et al., 2016; Geremew et al., 2016; Rundberget et al., 2004; Zhang et al., 2017) has attracted more and more consideration. One of the most important reasons is that these toxigenic fungi can naturally produce secondary metabolites-mycotoxins, which exhibit serious toxicities and considerable risks to the consumers (Edite Bezerra da Rocha et al., 2014). Among these mycotoxins, aflatoxins (AFs) mainly produced by *Aspergillus flavus* are toxic and carcinogenic metabolites, which have been classified as the Group IA carcinogen by the International Agency for Research on Cancer (IARC) (Al-Zoreky and Saleh, 2017; Iqbal et al.,

2017; Yang et al., 2017). OTA mainly produced by *Aspergillus carbonarius* (Kanapitsas et al., 2016) and *Penicillium viridicatum* (Bragulat et al., 2008) is reported to be a nephrotoxin, an immune suppressant, as well as a carcinogen, and is reported to be related with Balkan Endemic Nephropathy (BEN) of human (Grollman and Jelakovic, 2007; Vecchio et al., 2012; Vrabcheva et al., 2004). Therefore, over the last few decades, exploring for effective, safe, and economic candidates for controlling or prohibiting fungal contamination and mycotoxins residue in the above-mentioned matrices has got increasing focus and interests of researchers.

Physical and chemical preservatives such as ultraviolet and γ-rays irradiation, sulphur and aluminium phosphide fumigation, etc, have been introduced to inhibit the growth of fungi. Nevertheless, due to the potential decomposition reactions and residues, as well as their own potential toxicity, and potential undesirable biological effects on

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human health, these strategies have been forbidden in many official documents (da Cruz Cabral et al., 2013; Li et al., 2016; Prakash et al., 2015). Recently, some plant essential oils (PEOs) and their active components with strong and broad-spectrum antifungal properties as the succedaneums to protect foods, agricultural commodities and CHMs from fungal contamination in a safe, non-residual, biodegradable and environment-friendly way have been taken into consideration in some aspects (Prakash et al., 2012a,b; Viuda-Martos et al., 2011). Several studies have shown the antifungal and antimicrobial properties of PEOs from *Syringa* Linn., *Cinnamomum cassia* Presl, *Litsea cubeba*, *Cymbopogon martini*, *Foeniculum vulgare* Mill, etc, and they were of great value in foods, cosmetics, and cigarettes (da Cruz Cabral et al., 2013; Li et al., 2016; Soylu et al., 2006; Van Haute et al., 2016; Velluti et al., 2004; Zuzarte et al., 2013). It could be concluded that most of the reported bacteria or fungi were not toxigenic and could not produce mycotoxins. The relationships of the antifungal and antimicrobial properties and the internal compositions of PEOs were not explained. In addition, the antifungal effects of PEOs from the above-listed CHMs on the toxigenic fungi such as *Aspergillus flavus*, *Penicillium viridicatum* and *Aspergillus carbonarius* have not been clarified from then on. Therefore, evaluation on the antifungal activities of EOs from some widely-distributed and easy-to-obtain plants on the toxigenic fungi widely-found in the foods, agricultural commodities and other CHMs matrices that are easily to be contaminated by toxigenic fungi to explore natural and highly-effective mould inhibitor is still urgently needed.

As a proof of concept, this study aimed to firstly evaluate the inhibitory activities of 11 kinds of PEOs from anticipated candidates including *Cinnamomum cassia* Presl, *Litsea cubeba*, *Cymbopogon martini*, *Thymus mongolicus* Ronn, *Syringa* Linn., *Lavendula angustifolia* Mill., *Foeniculum vulgare* Mill, *Citrus reticulata* Banco, *Mentha haplocalyx* Briq., *Allium sativum* and *Artemisia argyi* on the toxigenic fungi of *Aspergillus flavus*, *Aspergillus carbonarius* and *Penicillium viridicatum* by using the modified *in vitro* inhibition zone assay through fumigation mode. Then, the minimal inhibitory concentrations (MIC) of the PEOs with stronger antifungal activities on the selected fungi were determined with the same method. The chemical compositions of PEOs with stronger antifungal activities were analyzed by an optimized gas chromatography-mass spectrometry (GC-MS) method. Then, the inhibitory activities of main active components in the selected PEOs were assessed and compared to elucidate the relationship between the active components and their antifungal properties. The schematic representation for this study was presented in Fig. 1. This work will provide powerful references for exploring highly-effective, safe, and economic antifungal agents

including PEOs together with some active compounds with low cost and wide source for inhibiting or controlling fungal contamination and mycotoxins residue in a large number of foods, agricultural commodities and CHMs matrices with high edible or medicinal values to assure their quality and safety for clinical application.

2. Materials and methods

2.1. Materials

Plant essential oils extracted from *Cinnamomum cassia* Presl, *Litsea cubeba*, *Cymbopogon martini*, *Thymus mongolicus* Ronn, *Syringa* Linn., *Lavendula angustifolia* Mill., *Foeniculum vulgare* Mill, *Citrus reticulata* Banco, *Mentha haplocalyx* Briq., *Allium sativum* and *Artemisia argyi* were purchased from Jiang Xi XueSong Natural Medicinal oil Co., Ltd. (Jiangxi, China).

Trans-cinnamaldehyde, citral, *trans*-geraniol, carvacrol d-limonene and citronellal were bought from Shanghai Macklin Biochemical Co., Ltd. (Shanghai, China).

The aflatoxigenic *Aspergillus flavus* lyophilized powder (CGMCC 3.4410) and *Penicillium viridicatum* lyophilized powder (CGMCC 3.4038) were purchased from the China General Microbiological Culture Collection Center (Beijing, China). And ochratoxigenic *Aspergillus carbonarius* (ITEM-5222) lyophilized powder was supplied by Agri-Food Toxigenic Fungi Culture Collection (ITEM) of the Institute of Science of Food Production, CNR (Bari, Italy). All the powders were dissolved in 0.5 mL of sterile water (121 °C, 20 min). *A. flavus* and *A. carbonarius* spores were cultured on the Salt Czapek Dox Agar medium (SCDA; Qingdao Hope Bio-Technology Co., Ltd., Qingdao, China) and *P. viridicatum* spore was cultured on the Potato Dextrose Agar Medium (PDA; Beijing Aoboxing Bio-Technology Co., Ltd., Beijing, China) at constant temperature and humidity (28 °C, 95% relative humidity (RH)) respectively.

2.2. Chemicals and reagents

Chromatographic-grade hexane was purchased from Thermo Fisher Scientific (Beijing, China). Other reagents and chemicals were all analytical grades and obtained from Beijing Chemical Works (Beijing, China).

The Czapek Dox Agar (CDA) medium prepared by dissolving 2 g NaNO_3 , 1 g K_2HPO_4 , 0.5 g MgSO_4 , 0.5 g KCl, 0.01 g FeSO_4 , 30 g sucrose, 15 g agar in 1000 mL of distilled water ($\text{pH} 6.8 \pm 0.2$) was purchased

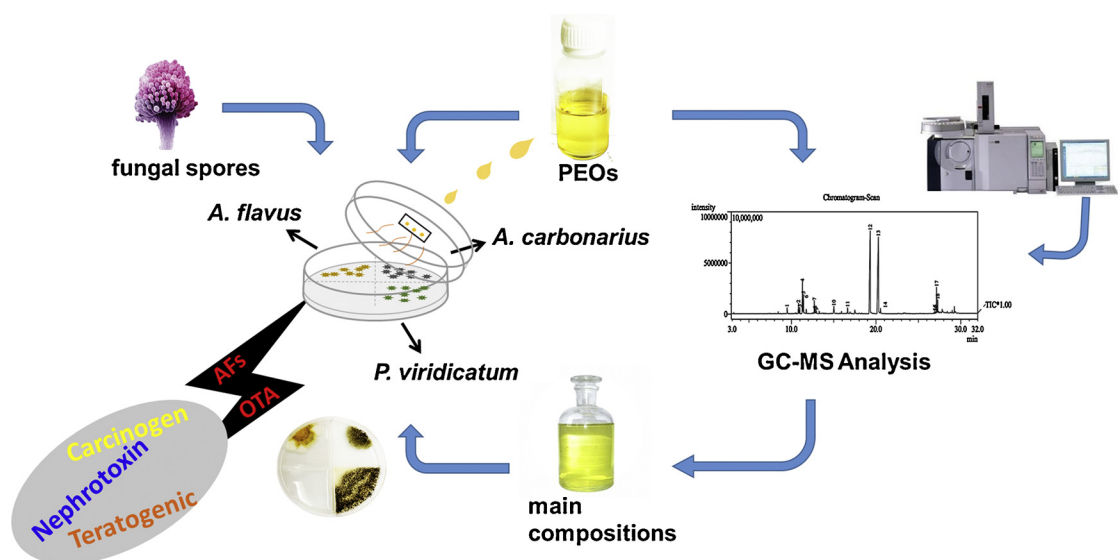


Fig. 1. Schematic representation for antifungal evaluation of plant essential oils and their main active components.

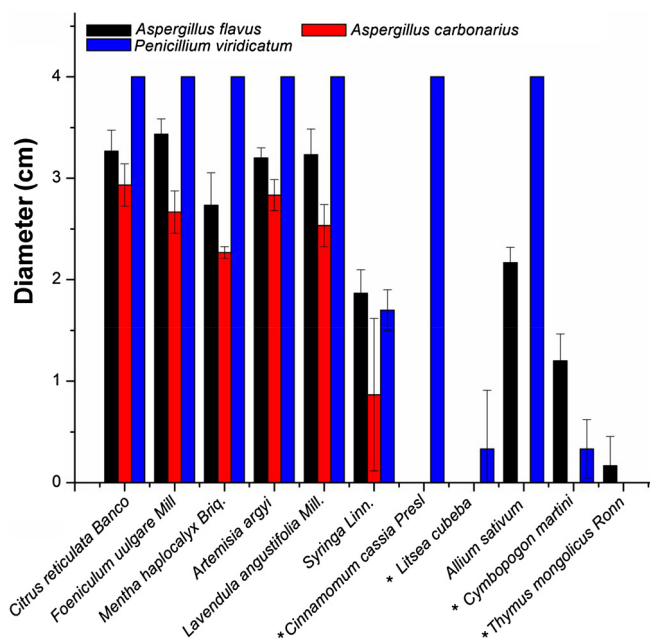


Fig. 2. Fungi zone diameters of *A. flavus*, *A. carbonarius* and *P. viridicatum* growth under the effect of 11 PEOs.

from Aobox Biotechnology Co., Ltd. (Beijing, China). Potato Dextrose Agar (PDA) medium, which was prepared by dissolving 200 g potato, 20 g glucose, 15–20 g agar in 1000 mL of distilled water (Natural pH) was purchased from Aobox Biotechnology Co., Ltd. (Beijing, China).

2.3. Preparation of fungi spores suspension

A. flavus and *A. carbonarius* were inoculated on the CDA medium, respectively, and *P. viridicatum* on the PDA medium, which were transferred into an incubator for cultivation for 7 days. Then, the fungi spores were suspended with 15 mL of sterile 0.01% tween-20 solution on the surface of CDA or PDA medium. Afterwards, the spores and mycelium mixture solution was removed by gravity filtration through at least three layers of autoclaved cheese cloth (Christensen et al., 2012). The concentration of spore suspension was adjusted to 10^7 colony forming units (CFU)/mL with a haemocytometer slide (0.1 mm depth, $1/400 \text{ mm}^2$) and an XDS-1 B optical microscope (Chongqing, China) to get the standard spore suspension for subsequent use (Li et al., 2016).

2.4. Antifungal evaluation of the 11 tested plant essential oils

For this assay, the modified inhibition zone method (Spooner et al., 1972) through fumigation mode based on transpiration of volatile odor was introduced. The three kinds of spore suspensions of *A. flavus*, *A. carbonarius* and *P. viridicatum* were inoculated on the center of CDA (for *A. flavus* and *A. carbonarius*) or PDA (for *P. viridicatum*) medium in the chassis of Petri dish, respectively, and cultured under the sterile environment. 10 μL of each PEO at the same concentration were dropped onto the surface of a fresh filter, which was pasted on the cover of the Petri dish. Then, the whole Petri dish was integrated with the chassis and the cover, and was sealed by the sealing film for incubation under the conditions of temperature 28°C and relative humidity 95% for 5 days. The zone diameter of fungi growth on the surface of culture media was determined to assess and compare the antifungal activities of the 11 tested PEOs, further to select PEOs with stronger antifungal activities. Based on the fumigation mode and its principle for inhibition zone, the smaller the zone diameter is, the stronger the antifungal activity the PEO exhibits.

Table 1

The MIC of various plant essential oils against *Aspergillus flavus*, *Aspergillus carbonarius* and *Penicillium viridicatum*. ($n = 3$).

Fungi	<i>Cinnamomum cassia</i> Presl ($\mu\text{L}/\text{mL}$) ^a	<i>Litsea cubeba</i> ($\mu\text{L}/\text{mL}$)	<i>Cymbopogon martini</i> ($\mu\text{L}/\text{mL}$)	<i>Thymus mongolicus</i> Ronn ($\mu\text{L}/\text{mL}$)
<i>Aspergillus flavus</i>	1.67	3.33	3.33	> 5.0
<i>Aspergillus carbonarius</i>	1.67	1.67	1.67	2.33
<i>Penicillium viridicatum</i>	> 5.0	3.33	> 5.0	> 5.0

^a $V_{\text{PEO}}/V_{\text{Medium}}$.

2.5. Minimal inhibitory concentration of the selected PEOs with stronger antifungal activities

The minimal inhibitory concentration (MIC) of the selected PEOs with stronger antifungal activities was measured by using the above-mentioned inhibition zone method through fumigation mode. Firstly, 10 μL of spore suspension of the tested fungi was inoculated on the CDA or PDA medium in the chassis of Petri dish with four aliquot parts (Fig. 1). Then, different concentrations of PEOs (1, 3, 5, 7, 10, 15 $\mu\text{L}/3 \text{ mL}$, $V_{\text{PEO}}/V_{\text{Medium}}$) were dropped onto a fresh filter pasted on the cover of the Petri dish to inhibit the growth of tested fungi. The integrated Petri dish were sealed by sealing film and incubated for 5 days under the conditions of temperature 28°C and humidity 95%. The growth of tested fungi on the surface of culture media was determined and the concentration of PEOs that completely inhibited the growth of fungi was decided as MIC.

2.6. GC–MS analysis of chemical composition of the selected PEOs

A GC-2010 gas chromatography (Shimadzu) equipped with GC/MS-QP2010 Plus mass spectrometer (Shimadzu) was used to analyze the chemical compositions of the PEOs with the strongest antifungal activities. 1.0 μL of PEOs dissolved in hexane was injected into the GC–MS system with a split ratio of 1:10 for the components separation on a fused silica capillary HP-5 column (30 m $\times$ 0.25 mm internal diameter, 0.25 μm film thickness). Helium was set as the carrier gas at a constant flowrate of 1.0 mL/min. The GC settings were as follows: initial column oven temperature was kept at 40°C for 3 min, and then was gradually ramped to 120°C at a rate of $5^\circ\text{C}/\text{min}$ and hold of 3 min, and then programmed to 280°C at $30^\circ\text{C}/\text{min}$ and hold for another 5 min. The MS operating parameters were listed as follows: injector temperature, 250°C ; detector temperatures, 250°C ; ion source temperature, 230°C ; ionization energy, 70 eV; total ion scan mode with mass scan range, 33–800 amu (scan time: 0.2 s). The qualitative identification of compositions in PEOs was based on the comparison of the mass spectra with standards recorded in the National Institute of Standards and Technology (NIST) libraries in combination with the published data. While, quantitative analysis of the individual component was carried out by peak area normalization measurement and the relative content of each component was calculated by $\% \text{peak area}$.

2.7. Antifungal evaluation of the main active components in the selected PEOs

It has been reported that the main active components such as *trans*-cinnamaldehyde and d-limonene in PEOs of *Cinnamomum cassia* Presl, citral and citronellal of *Litsea cubeba*, *trans*-geraniol of *Cymbopogon martini*, and carvacrol in *Thymus mongolicus* Ronn exhibited strong antimicrobial or antifungal activities (Miron et al., 2014; Shreaz et al., 2016; Umagiliyage et al., 2017; Xie et al., 2017; Zheng et al., 2015). But, the antifungal activities of them on the three kinds of toxigenic

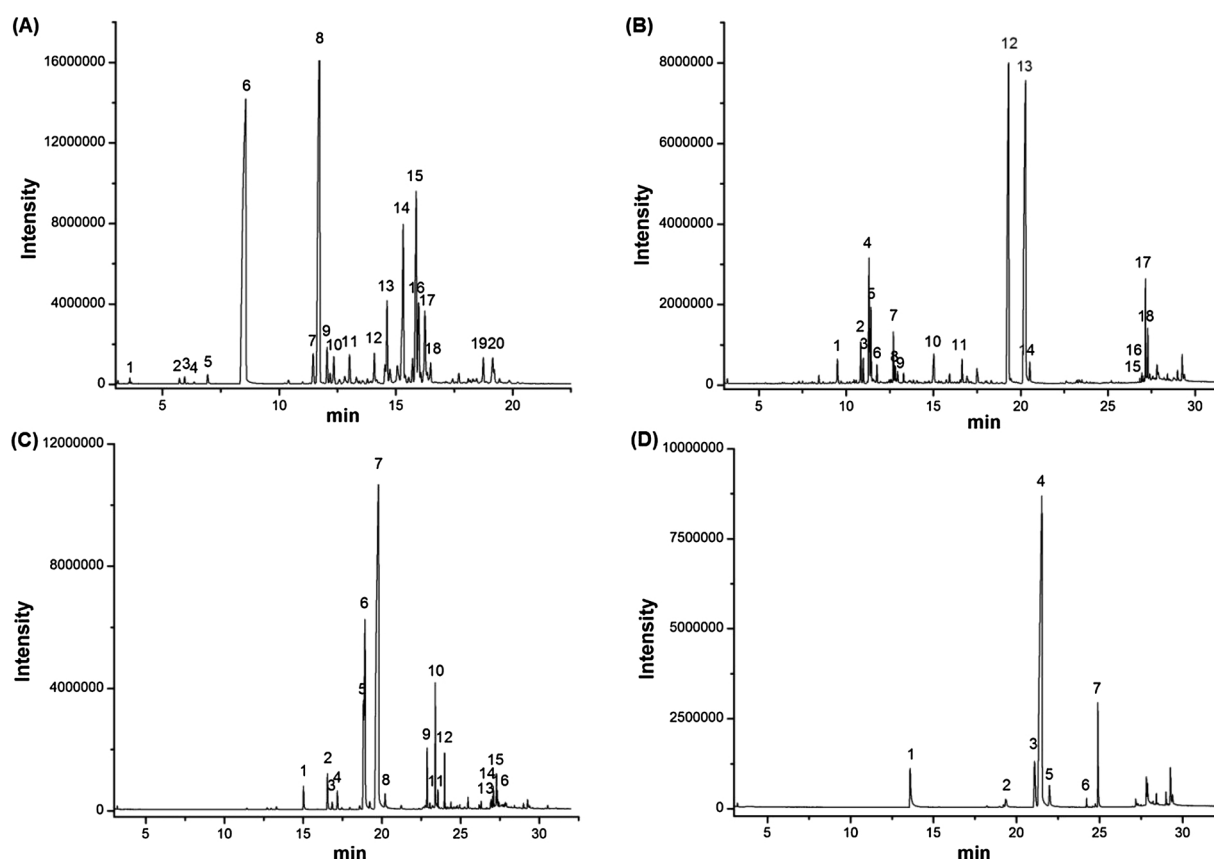


Fig. 3. GC-MS chromatograms of PEOs from (A) *Cinnamomum cassia* Presl, (B) *Litsea cubeba*, (C) *Cymbopogon martini* and (D) *Thymus mongolicus* Ronn.

Table 2
Chemical compositions of *Cinnamomum cassia* Presl essential oil.

Identified peaks	Retention time (min)	Formula	Compound	Relative content (%)
1	3.607	C ₁₀ H ₁₆	d-Limonene	0.13
2	5.727	C ₉ H ₁₀ O	Benzenepropanal	0.21
3	5.950	C ₁₀ H ₁₈ O	Borneol	0.20
4	6.361	C ₁₀ H ₁₈ O	3-Cyclohexene-1-methanol	0.07
5	6.934	C ₉ H ₈ O	Cinnamaldehyde	0.40
6	8.563	C ₉ H ₈ O	trans-Cinnamaldehyde	35.22
7	11.453	C ₁₅ H ₂₄	(+)-Cycloisotavene	1.40
8	11.722	C ₁₅ H ₂₄	Copaene	26.47
9	12.053	C ₁₅ H ₂₄	Cyclohexane	1.36
10	12.345	C ₁₅ H ₂₄	Humulen-(v1)	0.86
11	13.013	C ₁₅ H ₂₄	Caryophyllene	1.15
12	14.078	C ₁₅ H ₂₄	α-Caryophyllene	1.03
13	14.622	C ₁₅ H ₂₄	Naphthalene	3.33
14	15.309	C ₁₅ H ₂₄	α-Murolene	8.21
15	15.712	C ₁₅ H ₂₄	Isoledene	0.94
16	15.865	C ₁₅ H ₂₄	α-Cadinene	9.79
17	15.972	C ₁₅ H ₂₂	Calamenene	3.83
18	16.248	C ₁₅ H ₂₄	α-Cedrene	3.40
19	18.741	C ₁₅ H ₂₄	α-Longipinene	0.87
20	19.147	C ₁₅ H ₂₆ O	α-Cadinol	1.10

Table 3
Chemical compositions of *Litsea cubeba* essential oil.

Identified peaks	Retention time (min)	Formula	Compound	Relative content (%)
1	9.499	C ₁₀ H ₁₆	α-Pinene	1.09
2	10.836	C ₁₀ H ₁₆	β-Phellandrene	1.74
3	10.981	C ₁₀ H ₁₆	β-Pinene	1.13
4	11.307	C ₈ H ₁₄ O	Methyl heptenone	6.66
5	11.418	C ₁₀ H ₁₆	β-Myrcene	3.54
6	11.759	C ₁₀ H ₂₂	Decane	0.76
7	12.706	C ₁₀ H ₁₆	d-Limonene	2.49
8	12.807	C ₁₀ H ₁₈ O	Eucalyptol	0.79
9	12.949	C ₁₀ H ₁₆	3-Carene	0.53
10	12.022	C ₁₀ H ₁₈ O	Octadien	1.80
11	16.642	C ₁₀ H ₁₈ O	β-Citronellal	1.16
12	19.307	C ₁₀ H ₁₆ O	β-Citral	34.17
13	20.273	C ₁₀ H ₁₆ O	α-Citral	38.01
14	20.521	C ₇ H ₁₄	1-Pentene	1.35
15	26.938	C ₂₀ H ₃₄ O	Geranyl linalool	0.25
16	27.033	C ₁₅ H ₂₄	α-Farnesene	0.22
17	27.148	C ₂₀ H ₃₂	2,6,11,15-Tetramethyl-hexadeca-2,6,8,10,14-pentaene	2.79
18	27.284	C ₂₀ H ₃₄ O	(E,E,E)-3,7,11,15-Tetramethylhexadeca-1,3,6,10,14-pentaene	1.53

fungi including MICs have not been studied. Therefore, the antifungal activities of the 6 main active components on *A. flavus*, *A. carbonarius* and *P. viridicatum* were evaluated by using the modified zone method (Spoonster et al., 1972). Firstly, the spores suspensions of *A. flavus*, *A. carbonarius* and *P. viridicatum* were inoculated on the CDA or PDA medium in the petri dishes, respectively, then the main active components at different concentration levels (1, 3, 5, 7, 10, 15 μL/3 mL, $V_{\text{compound}}/V_{\text{Medium}}$) of the 6 compounds were dropped onto a fresh filter pasted on the cover of the Petri dish to inhibit the growth on the culture

media in chassis of Petri dish through fumigation mode. The sealed petri dishes were incubated in an incubator for 5 days at temperature 28 °C and relative humidity 95%. The MIC of each component on the three kinds of fungi was determined by observing the growth of fungi.

Table 4
Chemical compositions of *Cymbopogon martini* essential oil.

Identified peaks	Retention time (min)	Formula	Compound	Relative content (%)
1	15.024	C ₁₀ H ₁₈ O	β-Linalool	1.15
2	16.538	C ₁₀ H ₁₈ O	Isopulegol	1.94
3	16.839	C ₁₀ H ₁₈ O	Isopinocampheol	0.29
4	17.171	C ₁₀ H ₁₈ O	(–)-Isopulegol	0.99
5	18.829	C ₁₀ H ₁₈ O	cis-Geraniol	7.99
6	18.925	C ₁₄ H ₂₆ O ₂	Citronellyl butyrate	15.08
7	19.780	C ₁₀ H ₁₈ O	trans-Geraniol	58.98
8	20.198	C ₁₀ H ₁₆ O	α-Citral	0.81
9	2.866	C ₁₂ H ₂₂ O ₂	Citronellyl acetate	2.43
10	23.387	C ₁₀ H ₁₈ O	Geranyl acetate	5.24
11	23.559	C ₁₅ H ₂₄	β-Elementene	0.66
12	23.979	C ₁₅ H ₂₄	Caryophyllene	1.91
13	27.025	C ₁₅ H ₂₄	α-Caryophyllene	0.28
14	27.083	C ₁₀ H ₁₈ O	β-Citronellal	0.59
15	27.270	C ₁₂ H ₂₀ O ₂	Isopulegyl acetate	1.13
16	27.325	C ₂₀ H ₃₄ O ₂	cis-Geraniol	0.54

Table 5
Chemical compositions of *Thymus mongolicus* Ronn essential oil.

Identified peaks	Retention time (min)	Formula	Compound	Relative content (%)
1	13.593	C ₇ H ₈ O	o-Cresol	4.40
2	19.345	C ₁₀ H ₁₄ O	Thymol	1.08
3	21.079	C ₁₀ H ₁₄ O	Isothymol	6.94
4	21.522	C ₁₀ H ₁₄ O	Carvacrol	79.36
5	21.977	C ₁₀ H ₁₄ O	p-thymol	2.28
6	24.209	C ₁₃ H ₂₀ O	2,4-Diisopropyl-5-methylphenol	0.44
7	24.893	C ₁₂ H ₁₈ O	2,6-Diisopropylanisole	5.95

2.8. Statistical analysis

The experiments were performed in triplicate and the data were analyzed by one-way analysis of variance (ANOVA) of SPSS (version 21.0). $p < 0.05$ was considered as statistical significance.

3. Results and discussion

3.1. Antifungal evaluation of the 11 plant essential oils

The addition of 11 PEOs would affect or inhibit the growth of the tested fungi on the surface of culture media, which could be reflected from the fungi zone diameters. From the data of zone diameters of fungi growth under the effect of 11 kinds of PEOs at the same concentration in Fig. 2, it could be obviously found that *P. viridicatum* with bigger zone diameters grew better than *A. flavus* and *A. carbonarius*, indicating significantly different inhibitory effects of the 11 PEOs on them. All the PEOs showed the strongest inhibitory effects on *A. carbonarius* (fungi zone diameters less than 3.0 cm), followed by *A. flavus* (fungi zone diameters less than 3.5 cm) and *P. viridicatum* (fungi zone diameters less than 4.0 cm).

Among the 11 tested samples, PEOs of *Cinnamomum cassia* Presl,

Litsea cubeba, *Cymbopogon martini* and *Thymus mongolicus* Ronn expressed much stronger antifungal activities on the three kinds of fungi with smaller fungi zone diameters than the others ($p < 0.05$). They could completely inhibit the growth of *A. carbonarius* as no fungi zone diameters were determined. PEOs of *Cinnamomum cassia* Presl and *Litsea cubeba* could completely inhibit the growth of *A. carbonarius* and *A. flavus*, while PEO of *Thymus mongolicus* Ronn completely inhibit the growth of *A. carbonarius* and *P. viridicatum*. Therefore, the 4 kinds of PEOs from *Cinnamomum cassia* Presl, *Litsea cubeba*, *Cymbopogon martini* and *Thymus mongolicus* Ronn can be selected as the novel natural antifungal candidates for preventing fungal contamination and mycotoxins residues in foods, agricultural commodities and CHMs matrices. While, it could also be concluded that each PEO could not simultaneously inhibit the growth of *A. flavus*, *A. carbonarius* and *P. viridicatum* vigorously. So, in practice, combined use of the selected PEOs may be more effective.

3.2. MICs of the 4 selected PEOs based on antifungal assay

The above results have indicated the antifungal activities of the 11 PEOs and that of *Cinnamomum cassia* Presl, *Litsea cubeba*, *Cymbopogon martini* and *Thymus mongolicus* Ronn expressed much stronger inhibitory effects on three toxigenic fungi of *A. flavus*, *A. carbonarius* and *P. viridicatum*. Here, the MICs of the 4 selected PEOs were measured with the addition of 0.33, 1.0, 1.67, 2.33, 3.33, 5.0 μL/mL (V_{PEO}/V_{Medium}) of PEOs. The results in Table 1 presented that *Litsea cubeba* have the smallest MIC for three kinds of fungi among the 4 PEOs, followed by *Cinnamomum cassia* Presl, *Cymbopogon martini* and *Thymus mongolicus* Ronn. As for the fungi, the lowest MICs of the 4 PEOs for *Aspergillus flavus* and *Aspergillus carbonarius* were both 1.67 μL/mL (V_{PEO}/V_{Medium}), and 3.33 μL/mL for *P. viridicatum*. And the MICs of *Cinnamomum cassia* Presl, *Cymbopogon martini* and *Thymus mongolicus* Ronn for the three kinds of fungi were all bigger than 5.0 μL/mL. Regarding *Aspergillus carbonarius*, the 4 PEOs showed the smallest MIC (1.67 and 2.33 μL/mL), indicating the highest susceptibility of the fungi to them. For *P. viridicatum*, the 4 PEOs exhibited the biggest MICs (3.33 and > 5.0 μL/mL). *Litsea cubeba* with the smallest MIC expressed the strongest antifungal activities for the three kinds of fungi. These results were consistent with the above-listed data of fungi zone diameters. It also confirmed that the 4 selected PEOs with stronger inhibitory effects on the toxigenic fungi would be the preferred candidates as natural antifungal agents in practice.

3.3. Chemical compositions of the 4 selected PEOs by GC–MS

Here, chemical compositions of the 4 selected PEOs with stronger antifungal activities were analyzed by GC–MS. From Fig. 3, it could be seen that the chemical components in the 4 selected PEOs have got satisfactory separation under the above-listed chromatographic and MS conditions. The identified peaks, together with their compound names and relative contents in the PEOs of *Cinnamomum cassia* Presl, *Litsea cubeba*, *Cymbopogon martini* and *Thymus mongolicus* Ronn were listed in Tables 2–5, respectively.

Fig. 3(A) showed that all the chemical compositions in the essential oil of *Cinnamomum cassia* Presl were well separated within 20 min and a

Table 6
The MIC of the main compounds of the selected essential oils against *Aspergillus flavus*, *Aspergillus carbonarius* and *Penicillium viridicatum*.

Fungi	trans-cinnamaldehyde (μL/mL) ^a	Citral (μL/mL)	trans-geraniol (μL/mL)	Carvacrol (μL/mL)	d-limonene (μL/mL)	Citronellal (μL/mL)
<i>Aspergillus flavus</i>	0.33	1.67	> 5.00	1.00	> 5.00	> 5.00
<i>Aspergillus carbonarius</i>	1.67	1.67	0.33	0.33	> 5.00	> 5.00
<i>Penicillium viridicatum</i>	> 5.00	1.67	> 5.00	1.00	– ^b	> 5.00

^a $V_{compound}/V_{Medium}$.

^b No inhibitory effect on *P. viridicatum*.

total of 20 volatile compounds including borneol, *trans*-Cinnamaldehyde, d-limonene, copaene etc., were identified in Table 2. *trans*-Cinnamaldehyde was confirmed to be the main active component with the highest relative content (%area) of 35.22%, followed by copaene of 26.47%. Other components such as α -cadinene, α -muurolene, calamenene, α -cedrene, naphthalene accounted for the area percentage of 9.79%, 8.21%, 3.83%, 3.40%, 3.33%, respectively. And the area percentages of the rest volatile compounds were lower than 1%.

The compositions in essential oil of *Litsea cubeba* have got good separation in 30 min (Fig. 3(B)), and 18 volatile compounds including α -pinene, β -phellandrene, β -pinene, citral, and citronellal etc., were detected and shown in Table 3. It could be seen that citral including α - and β -citral was the main active component with the highest relative content (%area) 72.18%. Other main volatile compounds like methyl heptenone, β -myrcene, 2,6,11,15-Tetramethyl-hexadeca-2,6,8,10,14-pentaene, d-limonene, octadien, β -phellandrene took relative content of 6.66%, 3.54%, 2.79%, 2.49%, 1.80%, 1.74%, respectively.

Fig. 3(C) expressed that the chemical components in the PEO of *Cymbopogon martini* were well separated within 30 min and 16 volatile peaks were identified regarding their compound names, formula, as well as relative content (%area) in Table 4. *trans*-geraniol with the highest relative content of 58.98% was the main active component. Other main components contained citronellyl butyrate with relative content of 15.08%, followed by geranyl acetate (7.99%), *cis*-geraniol (5.24%), citronellyl acetate (2.43%), isopulegol (1.94%), caryophyllene (1.91%), β -linalool (1.15%), and isopulegyl acetate (1.13%). And the area percentages of the rest volatile components were lower than 1%.

As shown in Fig. 3(D), the chemical compositions in essential oil of *Thymus mongolicus* Ronn have got satisfactory separation within 30 min. Only 7 volatile peaks with compound names, formula and relative content (%area) were identified and listed in Table 5. Carvacrol was confirmed to be the main active component with the highest relative content of 79.36%. Other components included isothymol (6.94%), 2,6-diisopropylanisole (5.95%), o-cresol (4.40%), p-thymol (2.28%), thymol (1.08), and 2,4-Diisopropyl-5-methylphenol (0.44%).

The four main active components including *trans*-cinnamaldehyde, citral, *trans*-geraniol, carvacrol and other two volatile compounds like d-limonene, citronellal are widely known for their antifungal properties for various fungi (Shreaz et al., 2016; Zheng et al., 2015; Xie et al., 2017; Miron et al., 2014; Umagiliyage et al., 2017). But their inhibitory effects on *A. flavus*, *A. carbonarius* and *P. viridicatum* have not been explained, which were explored in the next part.

3.4. Antifungal effects of the main active components in the selected PEOs

The 6 main target compounds in the 4 selected PEOs of *Cinnamomum cassia* Presl, *Litsea cubeba*, *Cymbopogon martini* and *Thymus mongolicus* Ronn were selected to test their antifungal activity against *A. flavus*, *A. carbonarius* and *P. viridicatum*. Here, the 6 main compounds at concentration of 0.33, 1.0, 1.67, 2.33, 3.33, and 5.00 $\mu\text{L/mL}$ ($v_{\text{compound}}/v_{\text{Medium}}$) were added to determine their MICs to the tested fungi by observing the growth of fungi.

Results in Table 6 showed that the 6 compounds exhibited different antifungal activity against *A. flavus*, *A. carbonarius* and *P. viridicatum*. Carvacrol with small MIC values expressed the strongest inhibitory effects on the three kinds of fungi, followed by citral, *trans*-cinnamaldehyde, *trans*-geraniol, citronellal and d-limonene. *trans*-cinnamaldehyde, citral and carvacrol showed the best antifungal activity on *A. flavus* with MICs of 0.33, 1.67 and 1.00 $\mu\text{L/mL}$, respectively. The MIC for *A. carbonarius* was 0.33 $\mu\text{L/mL}$, indicating that 0.33 $\mu\text{L/mL}$ ($v_{\text{compound}}/v_{\text{medium}}$) of carvacrol and *trans*-geraniol could completely inhibit the growth of *A. carbonarius*. While, 5.00 $\mu\text{L/mL}$ of d-limonene and citronellal could not completely inhibit the growth of *A. carbonarius*, and the MICs were not obtained under the present conditions. As for *P. viridicatum*, the MICs regarding the 6 target compounds were high, which were all > 5.00 $\mu\text{L/mL}$ for *trans*-cinnamaldehyde, *trans*-geraniol

and citronellal. Interestingly, the MIC of d-limonene for *P. viridicatum* was not got as it did not show inhibitory effects on the fungi. The 6 main active components with strong inhibitory effects on *A. flavus* and *A. carbonarius* could be also chosen as the natural antifungal compounds for preventing fungal contamination in foods, agricultural commodities and CHMs matrices.

Taken the MIC values in Tables 1 and 6 into account, it could be found that the MICs of the 6 compounds were all lower than those of the 4 selected PEOs containing the target compounds. So, it could be concluded that the main active components (*trans*-cinnamaldehyde, citral, *trans*-geraniol, carvacrol) with high relative contents, as well as inhibitory effects against *A. flavus*, *A. carbonarius* and *P. viridicatum* might play important role and show large attribution for the antifungal activity of PEOs from *Cinnamomum cassia* Presl, *Litsea cubeba*, *Cymbopogon martini*, *Thymus mongolicus* Ronn containing them, respectively. However, PEOs from *Cinnamomum cassia* Presl, *Litsea cubeba*, *Cymbopogon martini*, *Thymus mongolicus* Ronn with wider sources, lower cost, and more efficient separation and fractionation will be the preferred choices as natural antifungal agents for preventing fungal contamination in foods, agricultural commodities and CHMs matrices in real application.

4. Conclusions

Contamination of toxigenic fungi and mycotoxins residue in foods, agricultural commodities and CHMs has posed serious threaten to human and animals' health and safety. Exploring for effective, safe, and economic antifungal agents for controlling or prohibiting fungi growth and contamination is in great need. Plant essential oils and their active components with the advantages of natural origin, low cost, strong and broad-spectrum antifungal properties, minimal residue have been the ideal candidate of researchers and operators.

In this study, essential oils from 11 natural plants, together with their active components were evaluated regarding the antifungal activities on three toxigenic fungi including *A. flavus*, *A. carbonarius* and *P. viridicatum* through the determination of fungi zone diameter. The results have demonstrated that 4 PEOs of *Cinnamomum cassia* Presl, *Litsea cubeba*, *Cymbopogon martini* and *Thymus mongolicus* Ronn out of the 11 candidates expressed stronger antifungal activities and lower MICs against the three kinds of fungi, which might be related or attributed to their main active components including *trans*-cinnamaldehyde, citral, *trans*-geraniol, and carvacrol, respectively. The 4 selected PEOs and their main active components has large possibility and wide application prospect to be used as the novel natural antifungal alternatives for preventing fungal contamination and mycotoxins residues in foods, agricultural commodities and CHMs matrices. More joyfully and expectantly, the use of PEOs with highly volatile components as the natural antifungal agents is environment-friendly, easy-to-obtain, convenient and highly-effective with minimal residues in protecting a large number of matrices from fungal contamination to assure their quality and safety. Further study will focus on extending the real application of the selected PEOs in practice.

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